

2020 AMC 10A Problem 15 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 15. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 15

Problem:

A positive integer divisor of $12!$ is chosen at random. The probability that the divisor chosen is a perfect square can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

Answer Choices:

- A. 3
- B. 5
- C. 12
- D. 18
- E. 23

Solution:

The prime factorization of $12!$ is

$$2^{10} \cdot 3^5 \cdot 5^2 \cdot 7 \cdot 11$$

This yields a total of

$$11 \cdot 6 \cdot 3 \cdot 2 \cdot 2 \text{ divisors of } 12!$$

In order to produce a perfect square divisor, there must be an even exponent for each number in the prime factorization. Note that the divisor can't have any factors of 7 and 11 in the prime factorization because there is only one of each in $12!$. Thus, there are

$6 \cdot 3 \cdot 2$ perfect squares.

(For 2, you can have 0, 2, 4, 6, 8, or 102 s, etc.)

The probability that the divisor chosen is a perfect square is

$$\frac{6 \cdot 3 \cdot 2}{11 \cdot 6 \cdot 3 \cdot 2 \cdot 2} = \frac{1}{22} \implies \frac{m}{n} = \frac{1}{22} \implies m + n = 1 + 22 = \boxed{(E) 23}.$$

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